Development Of A Multi-Modal Surveillance System Integrating Aerial Imaging And Signal-Based Detection For Enhanced Situational Awareness

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Abstract

Modern surveillance systems often struggle in low-light, smoke, fog, foliage-covered, and partially occluded environments where camera-only methods lose reliability. Radar sensing remains robust in such conditions, but radar-only systems usually lack semantic context for confident operator decisions. This project proposes a low-cost multi-modal aerial surveillance framework that combines UAV-based vision, 24 GHz millimetre-wave radar sensing, and edge artificial intelligence to deliver real-time fused situational awareness.

The proposed platform is built on an F450 drone equipped with a Raspberry Pi 4 companion computer, camera module, LD2410 radar, and GPS. On-board processing integrates YOLOv8-pose for human detection and posture estimation, ByteTrack for persistent multi-object tracking, and a Doppler–CFAR–MTI pipeline for radar-based motion analysis. A lightweight Multi-Layer Perceptron (MLP) fuses radar and vision features to generate a unified threat probability for each tracked target.

To improve operational usefulness beyond detection, a rule-based behaviour analysis module identifies events such as loitering, intrusion, abnormal movement, group formation, and person-down conditions. A Kalman-filter trajectory predictor estimates short-horizon motion for early warning. Telemetry, detections, and alert streams are transmitted via WebSocket and MQTT to a backend server linked to a war-room dashboard and an Augmented Reality (AR) mobile application for field personnel. An LLM-assisted query interface supports natural-language access to stored event records.

Performance is evaluated on a self-collected dataset by comparing vision-only, radar-only, and fused models using precision, recall, and F_1 score, along with latency and robustness metrics. The expected outcome is a practical, reproducible, and budget-friendly surveillance solution that improves threat awareness on commodity hardware and supports deployment in resource-constrained academic, industrial, and civil-safety settings.

Keywords: Multi-modal surveillance, sensor fusion, UAV, mmWave radar, edge AI.

Chapter 1

Introduction

Surveillance and situational awareness are now central to defence, public safety, disaster response, and industrial security. In most real deployments, systems still depend heavily on visible-spectrum cameras. This creates a major reliability gap because camera performance drops under low light, smoke, fog, dust, dense foliage, and partial occlusion. These are exactly the environments where robust monitoring is most critical.

At the same time, low-cost embedded computing, edge AI frameworks, and compact RF sensors have matured enough to support practical multi-modal surveillance prototypes. This creates an opportunity to combine visual and signal-based sensing in a unified architecture that is both technically effective and affordable for academic-scale deployment.

1.1 Problem Context in India

In the Indian context, surveillance challenges are common in border-adjacent regions, industrial campuses, and disaster-prone areas where visibility is frequently degraded. UAV-based systems are attractive because they can rapidly cover wide terrain, but camera-only UAV platforms are still vulnerable to environmental conditions.

India's Directorate General of Civil Aviation (DGCA) regulations also shape practical implementation. Any multi-sensor UAV prototype above nano category thresholds must comply with registration and operational-zone requirements. Therefore, a deployable research prototype must be designed not only for technical performance but also for regulatory feasibility and local hardware availability.

1.2 Proposed System Overview

The proposed system is an end-to-end multi-modal aerial surveillance framework with four connected components:

(1). **Drone sensing node:** Quadcopter with Raspberry Pi 4, camera, LD2410 24 GHz radar, and GPS.

(2). **Backend server:** WebSocket/MQTT aggregation, behaviour analysis, event storage, and trajectory prediction.

(3). **War-room dashboard:** Real-time map, live feed, alerts, and query interface.

On-device inference combines YOLOv8-pose for human detection, ByteTrack for multi-object tracking, and a Doppler–CFAR–MTI radar pipeline. A lightweight fusion MLP estimates a unified threat score per tracked target.

1.3 Open-Source Repositories Used

The project design is informed by two major open-source references:

- **PLFMRADAR [1]:** signal-processing concepts for Doppler/MTI/CFAR and radar workflow grounding.
- **RuView [2]:** streaming architecture, tracker patterns, and RF-aware sensing pipeline references.

These references are adapted to a low-cost UAV-centric implementation rather than being reused directly.

1.4 Scope and Limitations

This work focuses on a prototype-level demonstration under controlled field trials. The scope includes real-time detection, tracking, fusion-based threat scoring, behaviour alerts, and interface-level operational visualization. It does not claim full defence-grade deployment.

Current limitations include constrained sensing range from low-cost radar, dependence on self-collected dataset scale, and environmental variability in outdoor operation. Stretch goals such as thermal fusion and long-range telemetry are treated as optional enhancements beyond core objectives.

Chapter 2

Problem Statement

2.1 Context

Surveillance systems operate in critical environments such as borders, industrial facilities, disaster zones, and public spaces, where reliable detection and tracking are essential. Despite advancements in sensors and processing, most systems rely on a single sensing modality, creating a fundamental limitation in challenging environmental conditions.

2.2 Limitations of Existing Systems

Existing approaches exhibit four major limitations:

L1. Visual-Spectrum Dependency. Camera-based systems degrade under low light, smoke, fog, and occlusion. Thermal cameras partially address this but are costly, power-intensive, and still lack direct range and velocity estimation.

L2. Limited Semantics in RF Systems. Radar provides range and Doppler-based velocity information but lacks semantic understanding such as object type or posture. CSI-based methods (SpotFi [3], FarSense [4], Widar 3.0 [5]) enable sensing in controlled environments but are not suitable for aerial deployment.

L3. Architectural Fragmentation. Aerial platforms and ground-based sensors operate independently, preventing cross-modal fusion, unified tracking, and coherent situational awareness, increasing operator workload.

L4. High Cost and Closed Systems. Commercial multi-modal platforms (e.g., Anduril EagleEye [6]) are expensive and proprietary, limiting accessibility, extensibility, and academic reproducibility.

2.3 Problem Statement

Existing surveillance systems rely primarily on single-modality sensing and exhibit significant performance degradation in low-visibility and non-line-of-sight conditions. Radar-based systems provide kinematic information but lack semantic interpretation, while current architectures fail to integrate aerial and ground sensing into a unified framework. Furthermore, existing multi-modal solutions are cost-prohibitive and closed in design.

There is therefore a need for a modular, low-cost multi-modal aerial surveillance system that integrates UAV-based imaging with signal-based detection (24 GHz mmWave radar), performs on-board fusion using edge AI, and delivers a unified situational-awareness interface for both operators and field users.

2.4 Scope Boundary

The system targets detection of persons and vehicles at altitudes of 10–50 m and ranges up to 50 m. Evaluation is conducted using a self-collected dataset in controlled environments. The system is a prototype and excludes autonomous flight and offensive RF applications.

Chapter 3

Significance of the Study

3.1 Overview

The proposed multi-modal aerial surveillance system is significant across academic, industrial, and societal dimensions, offering both technical contributions and practical applicability.

3.2 Academic Significance

The core contribution is a radar–vision fusion Multi-Layer Perceptron (MLP) deployed on a Raspberry Pi 4, extending prior radar–camera fusion work [7, 8] to low-cost UAV platforms. The comparison of vision-only, radar-only, and fused models provides a reproducible benchmark.

The integration of YOLOv8-pose, ByteTrack, and Doppler–CFAR–MTI radar processing demonstrates real-time edge AI feasibility on embedded hardware, with insights into performance and resource constraints. A rule-based behavior-analysis engine detects events such as loitering, intrusion, abnormal motion, and person-down using fused kinematic features.

A stretch contribution includes micro-Doppler vital sign detection (breathing $\approx$ 0.3 Hz) using low-cost 24 GHz radar. Additionally, an LLM-based interface enables natural-language querying of a real-time detection database, contributing to LLM-assisted decision systems. Together, these components demonstrate that advanced sensing, reasoning, and decision-support capabilities can be achieved on commodity hardware, making the system both technically novel and practically deployable.

3.3 Industrial and Operational Significance

The proposed architecture enables a range of real-world applications by combining robust sensing, real-time inference, and intuitive user interfaces. Its ability to operate under degraded visibility conditions and provide actionable intelligence makes it suitable for deployment in both civilian and defence scenarios.

Table 3.1: Key operational use cases and enabling capabilities.

Use Case	Capability
Border / perimeter monitoring	Radar + vision fusion enables detection under occlusion and poor visibility.
Industrial surveillance	Behavior analysis detects loitering, intrusion, and group formation.
Disaster response	Radar penetrates smoke/debris; micro-Doppler detects survivors (stretch).
Field situational awareness	AR mobile app provides real-time threat visualization and alerts.
Operator analysis	LLM-based query interface enables fast database interrogation.

The system architecture (drone edge AI, backend, dashboard, AR app) is adaptable to diverse surveillance scenarios.

3.4 Societal Significance

Operating within a low-cost range (INR 10,000–20,000) and using open-source tools, the system enables accessible multi-modal surveillance for academic, civil defence, and industrial applications. By lowering the cost barrier, it allows smaller institutions, local authorities, and resource-constrained organisations to deploy advanced situational-awareness systems without reliance on expensive proprietary solutions. This contributes to the broader democratisation of surveillance technology and supports initiatives in smart infrastructure, disaster preparedness, and public safety.

3.5 Beneficiaries

Table 3.2: Primary beneficiaries.

Beneficiary	Benefit
Researchers	Multi-modal UAV sensing platform and benchmark dataset.
Students	Reference architecture for edge AI and sensor fusion projects.
Security agencies	Low-cost surveillance prototype for monitoring applications.
Disaster-response teams	Detection capability in smoke and debris environments.
Industrial operators	Affordable drone-based monitoring complementing CCTV.

Chapter 4

Literature Review

This chapter reviews representative research relevant to the proposed multi-modal aerial surveillance system. The discussion follows the style of paper-wise analysis and highlights how each prior work contributes to the problem space. The review is grouped into four themes: UAV vision-based detection, radar-based sensing, Wi-Fi CSI and passive RF sensing, and multi-modal fusion with edge artificial intelligence. The chapter concludes with a synthesis that directly motivates the proposed system architecture.

4.1 UAV-Based Aerial Imaging and Detection

Redmon et al. [9] introduced the YOLO detection framework and showed that unified single-stage object detection can achieve real-time speed while maintaining strong accuracy. This work established the baseline for many practical surveillance systems where low latency is critical.

Jocher et al. [10] advanced this line with YOLOv8 and demonstrated improved training stability, faster inference variants, and broader task support including pose estimation. The pose branch is especially useful for behaviour-level surveillance because it captures posture cues rather than only bounding boxes.

Wojke et al. [11] proposed DeepSORT, combining Kalman filtering and appearance embeddings for identity-preserving multi-object tracking. Their method significantly reduced identity switches compared to motion-only association.

Zhang et al. [12] introduced ByteTrack and showed that associating both high- and low-confidence detections improves trajectory continuity under partial occlusion. This is directly relevant to aerial scenes where targets are often intermittently visible because of viewpoint changes and clutter.

Although these works provide strong visual detection and tracking foundations, camera-only pipelines remain sensitive to adverse weather, illumination changes, smoke, and foliage occlusion. They also do not directly provide radar-like range-velocity cues, motivating the inclusion of complementary sensing.

4.2 Radar-Based Detection

Skolnik [14] and Richards [13] provide the theoretical basis for practical radar perception, including Doppler processing, Moving Target Indication (MTI), and Constant False Alarm Rate (CFAR) detection. These techniques remain central for extracting motion from cluttered environments.

Patole et al. [16] surveyed automotive radar pipelines and highlighted radar's robustness in poor visibility, especially when optical sensors degrade. Their review also emphasised the challenge of semantic classification using radar alone.

Santra and Hazra [17] demonstrated short-range mmWave radar for human motion and gesture understanding using micro-Doppler patterns. Their results confirmed that RF sensing can preserve detection capability in challenging visual conditions.

In practical low-cost systems, modules such as LD2410 make radar sensing accessible; however, radar-only pipelines still face limited spatial detail and weaker semantic interpretation than vision systems. This motivates feature-level fusion rather than single-modality operation.

4.3 Wi-Fi Channel State Information and Passive RF Sensing

Kotaru et al. [3] proposed SpotFi and demonstrated device-free localization from Wi-Fi CSI using fine-grained channel information. Their results established CSI as a viable passive sensing signal beyond communication.

Zhang et al. [4] presented FarSense and showed how CSI phase/amplitude fluctuations can be used for respiration estimation. This work is important because it demonstrates subtle physiological sensing without wearable devices.

Zheng et al. [5] introduced Widar 3.0 for cross-domain gesture recognition and improved generalisation across environments and users. Their study highlighted how domain adaptation is critical for robust RF sensing.

Recent open-source implementations such as RuView [2] show practical real-time CSI data pipelines, tracker integration, and edge-to-dashboard streaming patterns. Even when CSI is not used as the primary modality in outdoor UAV operation, these architectural and algorithmic patterns are valuable references for robust RF-aware system design.

Most CSI literature remains strongest in indoor/static deployments and is sensitive to multipath noise, which limits direct transfer to airborne outdoor surveillance. Therefore, CSI is treated in this work as conceptual and architectural guidance rather than the primary sensing channel.

4.4 Multi-Modal Sensor Fusion and Edge Artificial Intelligence

Khaleghi et al. [19] categorised multi-sensor fusion into early, feature-level, and decision-level strategies. Their taxonomy is used in this proposal to justify selecting feature-level fusion for balancing interpretability, computational cost, and performance.

Chadwick et al. [8] demonstrated that combining radar and camera cues improves distant-object detection reliability over camera-only baselines. Their findings support the argument that fusion is most beneficial when one modality weakens.

Nabati et al. [7] proposed CenterFusion and showed that radar-camera feature integration improves object detection quality and robustness in adverse scenarios. This work motivates the use of lightweight learned fusion in this proposal.

On the deployment side, open-source systems such as PLFM_RADAR [1] demonstrate accessible signal-processing workflows, while modern embedded hardware (for example Raspberry Pi-class devices) enables practical edge inference for low-cost prototypes.

Despite progress, most published multi-modal systems either depend on expensive sensor stacks or focus on ground-vehicle contexts rather than UAV-centric field deployment with unified operator interfaces.

4.5 Synthesis

The reviewed works collectively indicate four clear conclusions: (i) vision models offer strong semantic understanding but are vulnerable in degraded visibility; (ii) radar sensing is robust under poor visual conditions but weak in semantic interpretation; (iii) CSI research provides useful RF sensing and pipeline ideas, but mostly in indoor settings; and (iv) fusion improves reliability, yet low-cost UAV-native end-to-end implementations remain limited.

Therefore, the proposed project positions itself as a practical integration effort: combining UAV-based imaging, low-cost mmWave radar, edge AI fusion, behaviour analysis, and real-time dashboard/mobile interfaces into one reproducible surveillance framework suitable for resource-constrained academic and operational contexts.

Chapter 5

Proposed Methodology

5.1 System Architecture

The high-level system architecture is illustrated in Figure ??. Sensors on the drone feed the companion computer, which performs edge inference and streams data to the backend. The backend aggregates streams, runs behavior analysis, maintains a Structured Query Language (SQL) detection database, and broadcasts to the two operator interfaces.

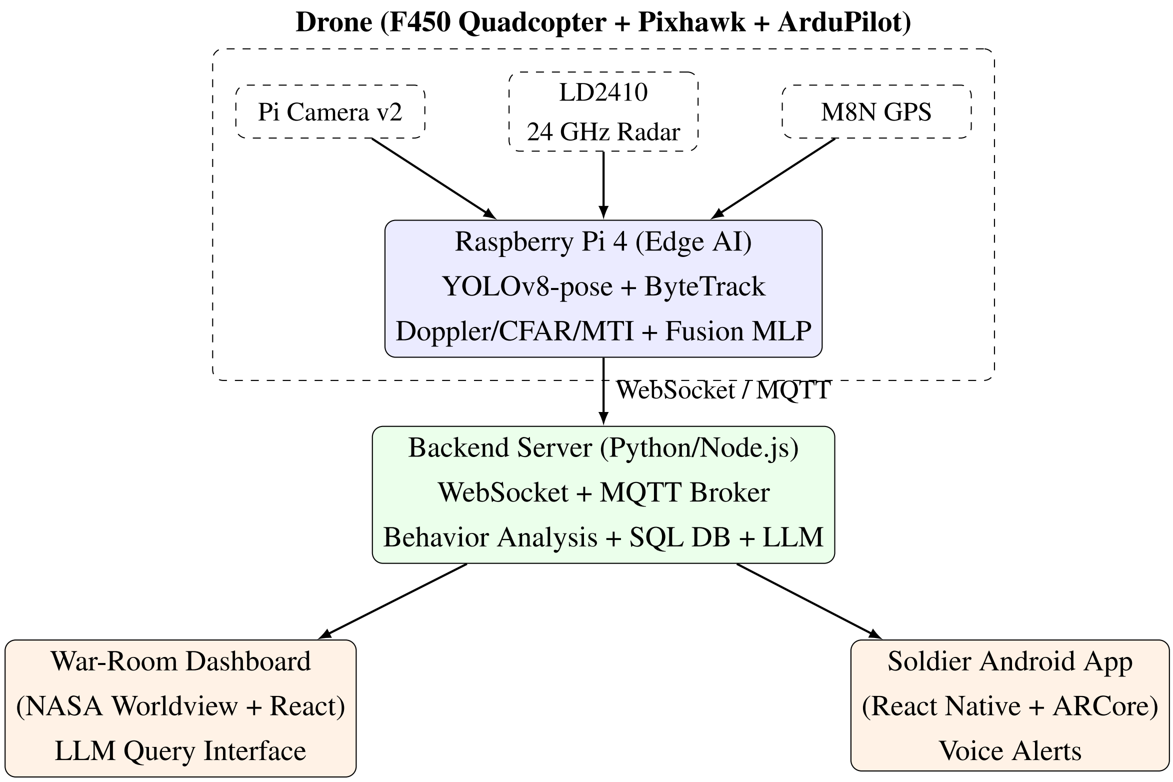

Figure 5.1: High-level system architecture of the proposed multi-modal aerial surveillance platform.

5.2 Hardware Subsystem

5.2.1 Airframe and Flight Controller

The drone is built around an F450 quadcopter frame carrying four 2212-class brushless motors controlled by four 30 A Electronic Speed Controllers (ESCs). A Pixhawk 2.4.8 flight controller running ArduPilot provides stable attitude control, GPS-based position hold, autonomous waypoint navigation, Return-To-Launch (RTL), and MAVLink telemetry. A 3S/4S 3000 mAh Lithium-Polymer (LiPo) battery provides flight power; a dedicated Battery Eliminator Circuit (BEC) supplies clean 5 V to the Raspberry Pi 4. All sensor mounts are 3D-printed from Polylactic Acid (PLA) using a friend's printer, keeping machining cost to zero.

5.2.2 Sensor Payload

The sensor payload consists of three units mounted below the frame in a fixed forward-looking configuration:

(1). **Raspberry Pi Camera Module v2** (8 MP, Sony IMX219): provides a 1080p visible-spectrum video stream at up to 30 fps, encoded as Motion JPEG (MJPEG) for low-latency streaming.

(2). **HiLink LD2410 24 GHz mmWave radar**: powered from the BEC, connected to the Raspberry Pi 4 over UART at 256,000 baud. The module outputs 13-byte basic data frames containing energy presence/movement flags and range data at up to 20 Hz, which are parsed and fed into the Doppler/CFAR/MTI processing chain.

(3). **u-blox M8N GPS module**: provides 10 Hz position fixes over UART to both the Pixhawk (for flight control) and the Raspberry Pi 4 (for telemetry tagging of detection events).

The overall hardware expenditure for the proposed multi-modal surveillance system is estimated to be in the range of **INR 9,800 to INR 11,700**, considering baseline components and typical market prices.

5.3 Software Stack

5.3.1 Drone-Side Software (Raspberry Pi 4)

A multi-threaded Python pipeline handles sensing, inference, and communication:

- **Camera**: captures Pi Camera v2 frames, encodes (MJPEG), publishes via MQTT.
- **Radar**: parses LD2410 data, applies MTI + CFAR, outputs detections.
- **Inference**: YOLOv8-pose (ONNX, quantised) + ByteTrack for tracking; fused with radar features via an MLP.
- **GPS**: streams M8N telemetry as GeoJSON at 10 Hz.

5.3.2 Backend Server

Implemented using FastAPI/Node.js with MQTT + WebSocket integration:

- Real-time data aggregation and broadcast.
- Behavior analysis and 5 s trajectory prediction (Kalman filter).
- SQL database for detection events.
- LLM query interface for natural-language analytics.

5.3.3 War-Room Dashboard (React + OpenLayers)

Provides real-time situational awareness:

- Live drone tracking, detections, and predicted paths.
- MJPEG video feed and alert panel.
- LLM-based query interface.
- Optional visual layers (ADS-B aircraft, satellite tracks, NVG/FLIR effects).

5.3.4 Soldier Android App (React Native + ARCore)

Delivers field-level intelligence:

- AR-based visualization of detections using GPS projection.
- Mini-map, compass guidance, and live alerts.
- Voice notifications via Android TTS.

5.4 Eight-Layer AI Stack

Table **??** summarises the full artificial-intelligence stack.

Table 5.1: Eight-layer AI stack with deployment location and implementation details.

Layer	Function	Location	Implementation
1	Person detection + posture	Drone (RPi 4)	YOLOv8-pose, ONNX Runtime, INT8; 17-keypoint COCO skeleton.
2	Multi-object tracking	Drone (RPi 4)	ByteTrack; Kalman prediction; persistent IDs.
3	Behavior analysis	Backend server	Rule engine: 5 alarm types; GPS geofence polygons.
4	Radar–vision fusion MLP	Drone (RPi 4)	6 features $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ 16 $\rightarrow$ sigmoid; $\sim$3,500 params.
5	LLM tactical query	Backend / Dashboard	Claude API + function-calling over SQL detection DB.
6	Trajectory prediction	Backend server	Kalman extrapolation 5 s; dotted path on map and AR.
7	(Stretch) Vital signs	Drone (RPi 4)	Micro-Doppler $\approx$0.3 Hz FFT on LD2410 close-range output.
8	Voice alerts	Soldier app	Android TTS; triggered by WebSocket alarm events.

5.5 Radar–Vision Fusion Model

5.5.1 Input Feature Vector

Let $\mathbf{x} = (x_1, x_2, x_3, x_4, x_5, x_6)^\top$ denote the feature vector for each active track, where:

- x_1 = radar Doppler velocity (m s^{-1}) from LD2410 processed output,
- x_2 = radar signal strength (arbitrary units, proxy for Radar Cross Section),
- x_3 = YOLO detection confidence $\in [0,1]$,
- x_4 = bounding-box area (fraction of frame area),
- x_5 = ByteTrack track duration (seconds since track initialisation),
- x_6 = ByteTrack track displacement rate (pixels per frame, converted to m s^{-1} using drone altitude and camera field-of-view geometry).

5.5.2 Model Architecture and Training

The fusion model is a three-hidden-layer fully-connected Multi-Layer Perceptron (MLP):

$$p(\mathbf{x}) = \sigma\Big(\mathbf{w}_4^\top\, \phi(\mathbf{W}_3\, \phi(\mathbf{W}_2\, \phi(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3) + b_4\Big) \tag{5.1}$$

where $\phi(\cdot) = \max(0, \cdot)$ is the Rectified Linear Unit (ReLU) activation, $\mathbf{W}_i \in \mathbb{R}^{d_i \times d_{i-1}}$ and $\mathbf{b}_i$ are trainable parameters with hidden dimensions d_1=64, d_2=32, d_3=16, and $\sigma(z) = (1+e^{-z})^{-1}$ is the sigmoid function producing a threat probability $p \in (0, 1)$.

The model has approximately 3,500 trainable parameters and is trained with binary cross-entropy loss and the Adam optimiser ($\alpha = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) for up to 200 epochs with early stopping on a held-out validation split (80 / 20 train/validation). Training data (≈500 samples) is collected across two to three field sessions, with positive labels assigned to human walking/running/crouching samples and negative labels to wind-driven foliage, animals, and vehicles. The trained model is exported to ONNX format for inference on the Raspberry Pi 4.

5.6 End-to-End Data Flow

Figure ?? summarises the end-to-end data flow from sensor capture to operator display.

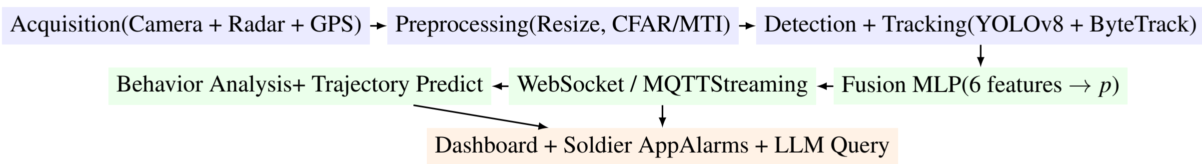

Figure 5.2: End-to-end data flow from sensor acquisition to operator output.

5.7 Evaluation Plan

Table ?? summarises the evaluation metrics, measurement method, and target values.

Table 5.2: Evaluation metrics, measurement methods, and targets.

Metric	Measurement Method	Target
Detection precision / recall / F_1	Vision-only vs. radar-only vs. fused MLP on held-out test split of field dataset.	Fused F_1 > both single-modality baselines.
Tracking ID switches	Count identity switches per minute on representative drone video clips.	<5 switches/min.
End-to-end alert latency	Timestamp difference between sensor frame capture (RPi 4) and alarm display on dashboard.	<500 ms (prototype).
Behavior alarm correctness	Confusion matrix over 5 alarm classes using annotated trial scenarios.	Precision >0.80 per class.
LLM query correctness	Manual review of LLM-generated SQL on a fixed 20-query battery.	≥85% correct.
Robustness under degraded visibility	F_1 delta between clear and degraded conditions (low light, partial foliage).	Fused model degrades less than vision-only.
RPi 4 CPU / RAM utilisation	`htop` and `psutil` logging during full-stack operation.	CPU <85%, RAM <1.8 GB.

Chapter 6

Research Gap Identification

6.1 Summary of Existing Work

The background surveyed in Chapter **??** reveals strong individual progress across four complementary domains:

- **Aerial imaging**: YOLOv8-pose delivers accurate, low-latency human detection with posture estimation; ByteTrack provides robust multi-object tracking across partial occlusions.
- **Radar detection**: CFAR, MTI, and Doppler-FFT algorithms are well-established for robust detection under clutter; low-cost 24 GHz mmWave modules bring these capabilities within a student budget.
- **Wi-Fi CSI sensing**: SpotFi, FarSense, and Widar 3.0 demonstrate device-free presence, motion, and pose estimation; the RuView project provides a production-grade software reference.
- **Multi-modal fusion**: feature-level MLP fusion consistently outperforms single-modality baselines in radar–camera systems; edge AI on Raspberry Pi 4 makes on-board fusion feasible.

Despite this progress, the integration of these techniques into a single, low-cost, reproducible airborne prototype with a unified operator interface remains largely absent from the published literature.

6.2 Identified Research Gaps

Table **??** maps each principal limitation of existing work to the specific gap it creates and the response proposed in this project.

Table 6.1: Mapping of existing limitations to identified research gaps and proposed responses.

Existing Limitation	Research Gap	Proposed Response
Camera-only UAV systems fail under smoke, fog, low light, and foliage occlusion.	No low-cost airborne platform combines visible-spectrum imaging with a penetrating RF modality.	Co-mount LD2410 24 GHz radar with Pi Camera on same F450 UAV; fuse streams on RPi 4.
Radar-only sensors lack semantic classification (human vs. vehicle vs. clutter).	No commodity airborne solution provides simultaneous Doppler detection and visual classification.	YOLOv8-pose for visual classification plus radar for Doppler velocity; MLP fuses both into a single threat score.
Wi-Fi CSI sensing literature is confined to indoor, fixed-node scenarios.	Passive RF sensing patterns are not integrated with aerial platforms or geospatial fusion.	Use RuView algorithmic patterns (Kalman tracker, WebSocket API) as software reference; apply CSI concepts to radar signal processing on UAV.
UAV imaging and ground sensors operate in isolated architectures with separate operator interfaces.	No unified end-to-end architecture from drone sensor to operator dashboard to soldier AR app exists in open-source literature.	Four-component architecture (drone, backend server, war-room dashboard, soldier AR app) connected by WebSocket and MQTT.
PLFM_RADAR and RuView exist as separate, domain-specific projects.	No combined design applies the algorithmic strengths of both inside one surveillance framework.	Adapt PLFM_RADAR Python CFAR/MTI scripts for LD2410; adopt RuView streaming API patterns for backend.
Behavior analysis (loitering, breach, group formation) on edge hardware is rarely demonstrated end-to-end.	Rule-based behavior engines over fused multi-modal tracks lack published low-resource implementations.	Behavior-analysis engine operating on ByteTrack and radar fused tracks; deployed on backend server.
Commercial multi-sensor UAV platforms cost many lakhs of rupees.	No budget-constrained multi-modal aerial surveillance reference design within INR 20,000 exists.	Full prototype with documented Bill of Materials targeting INR 10,000 to INR 20,000.

6.3 Central Research Gap Statement

On the basis of the analysis above, the central research gap is stated as follows:

> *Current surveillance systems commonly operate as isolated camera, radar, or wireless-sensing platforms. There is limited practical integration of UAV-borne aerial imaging, low-cost millimetre-wave radar signal-based detection, and edge artificial intelligence within a single open-source surveillance framework that is reproducible on commodity hardware, is formally evaluated against both single-modality baselines, and exposes a unified operator interface comprising both a war-room geospatial dashboard and a soldier-facing Augmented Reality application. This project addresses that gap by proposing a modular multi-modal aerial surveillance architecture in which a radar–vision Multi-Layer Perceptron fusion model is the principal academic contribution.*

6.4 Why the Gap is Researchable

The identified gap satisfies three criteria for a researchable problem:

(1). **Feasibility.** All hardware components are procurable through Indian distributors (Robu.in, Amazon.in) within the stated budget. All software components — YOLOv8-pose, ByteTrack, ONNX Runtime, ArduPilot, NASA Worldview, React Native, Google ARCore, FastAPI, and the Claude API — are freely available under open-source or free-tier licences.

(2). **Evaluability.** The primary academic claim — that the fused MLP detector outperforms vision-only and radar-only baselines — is directly testable using precision, recall, and F_1 score computed over a self-collected field dataset of approximately 500 labelled samples. Additional metrics (tracking continuity, alert latency, LLM query correctness) are also operationally measurable.

(3). **Novelty.** The specific combination of low-cost airborne mmWave radar with YOLOv8-pose fusion, evaluated on aerial data collected by the project team, has not been published to the team's knowledge at the time of proposal submission. The micro-Doppler vital-signs stretch objective adds a further novel dimension.

6.5 Boundary Between Gap and Scope

The project does *not* claim to address the following: (i) long-range active radar coverage beyond the LD2410's practical sensing envelope (approximately 5–10 m for human detection), (ii) night or thermal imaging (thermal camera is a stretch item), (iii) autonomous swarm behaviour, or (iv) encrypted or secure communications.

Chapter 7

Objectives

7.1 Main Objective

To design, develop, and evaluate a low-cost, modular, multi-modal aerial surveillance system that integrates UAV-borne visible-spectrum imaging with 24 GHz millimetre-wave radar-based detection, fuses both streams using edge artificial intelligence on a Raspberry Pi 4 companion computer into a unified threat-probability score, performs rule-based behavior analysis on the fused tracks, and presents the output through an operator war-room dashboard and a soldier-facing Augmented Reality mobile application — thereby demonstrating enhanced situational awareness over either single-modality baseline in an empirically evaluated, reproducible, open-source prototype.

7.2 Specific Objectives

O1. To study and analyse state-of-the-art techniques in UAV-based aerial detection (YOLOv8-pose), multi-object tracking (ByteTrack), millimetre-wave radar signal processing (CFAR, MTI, Doppler-FFT), passive Wi-Fi CSI sensing (RuView reference), and multi-modal feature-level fusion, including a detailed review of the algorithmic patterns available in the PLFM_RADAR and RuView open-source projects.

O2. To design the four-component system architecture comprising (i) an airborne F450 quadcopter with multi-sensor payload and Raspberry Pi 4 companion computer, (ii) a backend WebSocket and MQTT server, (iii) a war-room dashboard on top of NASA Worldview (React 19 + OpenLayers 10.9), and (iv) a soldier-facing Android application in React Native with Google ARCore, establishing all interface contracts (data formats, message schemas, WebSocket event types) before implementation begins.

O3. To implement the physical drone including the F450 frame assembly, 2212 brushless motors, 30 A Electronic Speed Controllers (ESCs), Pixhawk 2.4.8 flight controller with ArduPilot, 3S/4S LiPo battery system, and 3D-printed sensor mounts for the Raspberry Pi Camera Module v2 and LD2410 radar, followed by bench validation of each sensor individually and hover-stability testing.

O4. To implement the on-drone edge-inference stack including YOLOv8-pose human detection with 17-keypoint Common Objects in Context (COCO) posture estimation, ByteTrack multi-object tracking with persistent identifiers (T-01, T-02, ...) and Kalman-filter position prediction, and a Doppler–Constant False Alarm Rate (CFAR)–Moving Target Indication (MTI) signal-processing chain on the LD2410 UART serial output.

O5. To design and implement the radar–vision fusion Multi-Layer Perceptron (MLP) — a three-hidden-layer fully-connected network (64→32→16 units, ReLU activations, sigmoid output) trained on a self-collected field dataset of approximately 500 labelled samples (positive: humans walking/running/crouching; negative: wind-driven foliage, animals, vehicles) to produce a threat probability $p \in [0, 1]$ from six input features: radar Doppler velocity, radar signal strength, YOLO confidence, bounding-box area, track duration, and track displacement rate.

O6. To implement the backend behavior-analysis engine operating on fused multi-modal tracks to raise colour-coded alarms for:

- Loitering: same track identifier in a zone for more than N seconds (Yellow).
- Perimeter breach: track enters a GPS-defined restricted polygon (Red).
- Running: track displacement per frame exceeds a threshold (Orange).
- Group formation: three or more track identifiers converge within a defined radius (Yellow).
- Person-down: track keypoints collapse to a horizontal plane (Medical).

Along with a Kalman-filter trajectory predictor that extrapolates target position 5 seconds forward, rendered as dotted predicted paths on both the war-room map and the soldier AR view.

O7. To implement the war-room dashboard on top of the NASA Worldview React/OpenLayers codebase, adding: (i) a drone GPS track as a vector layer on satellite imagery, (ii) detection-event markers with threat-probability colour coding, (iii) a live MJPEG camera-feed panel, (iv) audio–visual alarms with severity thresholds, (v) an optional Night-Vision-Goggle (NVG)/Forward-Looking Infrared (FLIR) OpenLayers canvas shader for demo effect, and (vi) a Large Language Model (LLM) tactical query interface powered by the Claude API with function-calling over the SQL detection database.

O8. To implement the soldier-facing Android application using React Native and Google ARCore, providing: (i) live MJPEG drone camera feed, (ii) AR detection markers overlaid on the device camera view, (iii) a tactical mini-map showing drone and soldier positions, (iv) real-time threat alert notifications matching the war-room alarms, (v) a compass bearing to the nearest detected threat, (vi) drone battery level and status indicators, and (vii) Text-to-Speech voice alerts.

O9. To evaluate the system using the following metrics:

- Detection: precision, recall, F_1 score for vision-only, radar-only, and fused-MLP detectors on the field-collected test split.
- Tracking: identity-switch count per minute and track continuity across representative clips.
- Latency: end-to-end time from sensor frame capture to war-room alarm display (target $<$500 ms prototype, $<$250 ms final).
- Behavior analysis: confusion matrix over five alarm classes.
- LLM interface: query-correctness rate on a fixed 20-query evaluation battery.
- Robustness: F_1 score degradation between clear and degraded-visibility (low-light, partial foliage) conditions.

O10. To release all project outputs as an open-source repository including: drone firmware (ArduPilot configuration), RPi 4 edge-inference code (Python), backend server code (Python/Node.js), war-room dashboard source (React/JavaScript), soldier AR app source (React Native), fusion-MLP model weights (ONNX), field dataset (annotated JSONL), Bill of Materials, wiring diagrams, and 3D-printable sensor-mount STL files.

O11. (Stretch) To investigate radar micro-Doppler vital-signs detection from an airborne platform using the LD2410 at close range ($<$5 m), targeting the detection of stationary but breathing humans (breathing frequency $\approx$ 0.3 Hz micro-Doppler signature) behind light foliage or obstacles where YOLOv8-pose yields no detection, and to compare performance against the RuView Wi-Fi CSI vital-signs extraction approach as a conceptual baseline.

Chapter 8

Work Plan

8.1 Overview

The project spans twelve months across two semesters (Project-I: Months 1–5; Project-II: Months 6–12) and is divided into seven phases. Each phase produces a verifiable deliverable, ensuring early hardware validation and incremental software integration, with field evaluation beginning after core system validation.

8.2 Project-I (Months 1–5)

8.2.1 Phase 1: Drone Build and Sensor Validation (Months 1–3)

Activities:

- Procure hardware (motors, ESCs, Pixhawk, battery, LD2410, GPS, Pi Camera).
- Assemble F450 frame and power system; configure Pixhawk (ArduPilot).
- Calibrate ESCs, accelerometer, and compass.
- Validate sensors: camera (Picamera2), radar (UART 256000 baud), GPS.
- Mount sensors using 3D-printed fixtures.
- Perform tethered hover tests and GPS position hold.
- Complete DGCA registration if required.

Deliverable: Airworthy drone with validated sensors and stable hover.

8.2.2 Phase 2: Edge AI and Radar Processing (Months 4–5)

Activities:

- Deploy YOLOv8-pose (INT8, ONNX) on RPi 4 ($\geq$3 fps).
- Integrate ByteTrack for tracking.
- Implement CFAR/MTI/Doppler radar pipeline using LD2410 data.
- Develop multi-threaded pipeline (camera, radar, inference, GPS).

- Design MQTT schema and test streaming to backend.

Deliverable: Working edge AI + radar pipeline with verified MQTT stream and performance metrics.

8.3 Project-II (Months 6–12)

8.3.1 Phase 3: Backend and Behavior Analysis (Months 6–7)

Activities:

- Develop FastAPI/MQTT backend and WebSocket streaming.
- Implement SQL database and behavior rules (loitering, breach, etc.).
- Add Kalman trajectory prediction (5 s ahead).
- Test live drone-to-backend streaming and alarms.

Deliverable: End-to-end system with behavior detection and database logging.

8.3.2 Phase 4: Fusion MLP and Evaluation (Months 7–8)

Activities:

- Collect ∼500 field samples (positive/negative scenarios).
- Train six-feature MLP (Adam, BCE loss, 80/20 split).
- Evaluate vision-only, radar-only, and fused models (precision, recall, F_1).
- Analyse robustness under degraded conditions.

Deliverable: Trained ONNX MLP and benchmark evaluation report.

8.3.3 Phase 5: War-Room Dashboard (Months 8–9)

Activities:

- Extend NASA Worldview (React + OpenLayers).
- Add telemetry layers, detection markers, trajectories, MJPEG feed.
- Implement alarm panel and LLM query interface.
- Integrate external layers (OpenSky, CelesTrak).

Deliverable: Functional dashboard with real-time visualization and LLM queries.

8.3.4 Phase 6: Soldier Mobile Application (Months 10–11)

Activities:

- Develop React Native + ARCore app.
- Implement AR overlays, mini-map, compass, and TTS alerts.
- Test multi-device real-time synchronization.

Deliverable: Working AR-based mobile application.

8.3.5 Phase 7: Integration and Final Evaluation (Month 12)

Activities:

- Perform full system integration and field trials.
- Evaluate metrics (latency, F_1, tracking, robustness).
- Prepare report, demo video, and research paper.
- Release open-source code, dataset, and models.

Deliverable: Fully integrated prototype, evaluation results, and open-source release.

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